Chem 2310-001, Organic Chemistry University of Utah, Spring 2025

Version: Jan. 8th 2025

Class Details:
Instructor: Prof. Ming Lee **Tang** Credits: 4.00 (Letter grade)
Lecture: MWF 09:40 AM − 10:30 AM @ HEB 2004

- Lectures: will be recorded on Zoom, available the same afternoon via Canvas

Co- and Pre-requisites: a C- grade or better in Chem 1220 or Chem 1221 or AP Chemistry score of 5+
Discussion: Tuesdays (REQUIRED):

- 12:55 PM − 1:45 PM: Section 002: @ HEB 2006
- 9:40 AM − 10:30 AM: Section 003: @ CSC 208
- Ask questions/ practice Homework or Quiz material and get instant, in-person feedback from myself or the TAs

Exams during Spring 2025:

- **Exam 1**: Fri., Feb. 7th; **Exam 2**: Wed., Mar. 5th; **Exam 3**: Wed. Apr. 2nd; **Final**: Fri., Apr. 25th 8.00–10.00am.
- Calculators allowed but no help from anyone else is allowed.
- A photo-identification (Driver's license or Student ID) will be required
- Gradescope (entry code: **G3RKRK**) will be used for Quizzes, Discussion Worksheet, three Exams, Final and all regrade requests.

Grading:

- 15% **Exam 1**; 17% **Exam 2**; 18% **Exam 3**; 30% **Final**
- 3% **Participation**
    - 2% is for Poll Everywhere questions during class time/ synchronous lectures starting Friday, Week 2.
        - Canvas > Modules > Poll Everywhere (click to access)
        - Participate via browser/ phone: https://pollev.com/mingleetang668
        - Use the same email in Canvas for signing into Poll Everywhere to get credit
        - Full credit will be awarded for getting the right answer in at least 70% of the Polls.
        - Partial credit if your Poll Everywhere score is less than 70%: (Proportion of correct answers)×3, e.g. if you get 50% correct, your score is 0.5×3=1.5% out of 3%
    - 1% is reserved for attendance for the Discussions on Tuesdays. You can skip two Discussions with no penalty.
- 9% **Discussion Worksheet** (see timetable on Pg. 3)
    - This worksheet will be available after class on Mondays starting Week 2.
    - You'll work in fixed groups with an assigned Teaching Assistant during the mandatory Tuesday Discussions.
    - This worksheet is due on Gradescope on Wednesdays at 9:30 am before lecture.
    - This is to ensure you're abreast of the new concepts we've covered during lecture
    - The remainder 8% will reflect your score on the Discussion Worksheets. The 2 lowest scores will be dropped.
    - This will be graded by the Learning Assistants and Teaching Assistants.
- 8% **Quizzes** (see timetable on Pg. 3)
    - Take the quiz on Gradescope starting from Week 2
    - Quiz will be released after Friday's lecture and must be completed before noon the following Friday.
    - Quiz material will draw on concepts covered in the prior or current week, e.g. Quiz 1 due 11:59 pm Friday Week 2 will be released after Friday's lecture Week 1 and cover material presented during Week 1.
    - Your 8 best scores for the 10 quizzes will be taken into consideration for your grade.
    - This will be graded by the Teaching Assistants and myself; feedback will be given.
- Letter grades are based on class statistics (mean & standard deviation). It will reflect a grade distribution similar to other UoU instructors for this same class.
- MISSED EXAMS ARE NOT SUBJECT TO MAKE-UP; VALID EXCUSES WILL ALLOW PRO-RATING OF ONE MISSED MIDTERM (EXAM I, II OR III, BUT NOT all three); THE FINAL EXAM MUST BE TAKEN AS SCHEDULED.

Textbook:

- We will use David Klein's "Organic Chemistry", 3rd edition, Wiley, 2017.
    - For access, sign into Canvas and click on our Chem2310-001 Spring 2024 Bookshelf:
    - This ebook is offered via Inclusive Access in other sections of this class and will be used in the second part of

this series (Organic Chemistry II, not this class but the next one).
- It is important to choose the book that fits your style of learning. An alternative is "Smith, J.G. Organic Chemistry (4th or 5th edition); McGraw Hill". The e-book is available from several sites.

Canvas Discussion Board:
We will be using the Canvas Discussion Board. Rather than emailing class related questions to the teaching staff, please post your questions on Canvas. I will only answer class related questions via Canvas so everyone can have equal access to my response. This includes any chemistry related questions related to the Homework or Quizzes. Active engagement will boost your Participation score (worth 3% of your grade)

Interaction outside lecture:
- Email: minglee.tang@utah.edu. Please email me if it's something personal that requires individual attention.
- Office hours:
  - In-person on Tuesdays: 3:00–4:00 pm and Fridays: 12:00-1:00 pm in HEB 4259 (diagonally opposite Organic Labs).
- Announcements: Canvas will be used to deliver any new information you need to know.
- Learning Center Supplementary Instructor (SI): Claire Massouh
  - The SI is a student who has taken this course before and will facilitate two 60-minute group review sessions each week outside of class time.
- Center for Science and Math Education Learning Assistants (LAs): Natalia, Issac, Erika and Marlon
- Teaching Assistants (TAs):
  - Graduate TA: Shadi
  - Undergraduate TAs: Cathleen, Kyle, Adrian, Clarissa

Office Hours:

Mondays	Tuesdays	Wednesdays	Thursdays	Fridays
Natalia 11:00 am – 12:00 pm Curie Active learning center	Adrian 10:45 – 11:45 am 2130H (Level 2 of the library)	Erika 8:20 – 9:20 am Curie active learning center	Shadi 10:00 – 11:00 am HEB 4259	Cathleen 10:45 – 11:45 am HEB 4259
Isaac 12-2 pm Curie Active learning center	Natalia 11:45–12:45pm Curie active learning center	Kyle 1:30 – 3:30 pm Curie active learning center	Marlon 11:00 am – 12:00 pm Curie active learning center	Ming Lee 12:00 – 1:00 pm HEB 4259
	Ming Lee 3:00 – 4:00 pm HEB 4259		Clarissa 2:00 – 4:00 pm Curie Active learning center	
	Isaac 7:00 – 8:00 pm Zoom		Marlon 5:00 – 6:00 pm Zoom Meeting ID: 773 0240 4243 Passcode: QcZ4X5	

SI Sessions:

Mondays	Thursdays
12:15 PM - 1:15 PM	12:50 PM - 1:50 PM
HEB 2010	HEB 2006

Learning outcomes:
- Compare/predict the reactivity and stability of organic molecules based on their structure (hybridization, geometry, relevant **resonance** structures, atom size/electronegativity, charge).
- Use structure (ARIO) to predict acid/base strength and position of acid/base equilibrium.

- Determine structure of an organic molecule using the molecular formula, IR, MS, and NMR data.
- Predict curved arrow mechanisms of organic reactions.
- Predict products of an organic reaction based on the mechanism.
- Design syntheses of organic molecules.

Improper behavior:

- Any form of improper behavior during lecture or on examinations, will not be tolerated and will be forwarded to College of Science Academic Affairs Committee for investigation.
- For info on the University of Utah's policies regarding student conduct, see Student Code (Policy 6-400, Section V)
- Improper behavior includes (but is not limited to) cheating (including bringing unauthorized materials into an exam), marking a graded exam prior to a regrade request, class disruption, submitting another person's work as your own, sabotaging or otherwise interfering with another student's work, willful disobedience of directives from a teaching assistant, or removing materials from or bringing unauthorized materials into the lab. **Students engaged in improper behavior are subject to immediate dismissal from the course with a failing grade**.
- All study materials in this course are considered intellectual property of the instructor and the University of Utah (this includes lecture slides, problem sets, exams, answer keys etc.). *Unauthorized uploading or distribution of the aforementioned materials to any website, either during or after the semester, is prohibited and may be addressed both as a violation of the behavioral standards as well and an act of academic misconduct.* The Department will actively monitor websites for unauthorized distribution and refer all instances of the violation to VP for Student affairs and the College of Science Academic Affairs Committee for investigation.

University Policies

University Safety Statement. The University of Utah values the safety of all campus community members. To report suspicious activity or to request a courtesy escort, call campus police at 801-585-COPS (801-585-2677). You will receive important emergency alerts and safety messages regarding campus safety via text message. Lauren's Promise – I will listen and believe you if you are being threatened. If you are in immediate danger call 911. Utah Domestic Violence Coalition is another good resource 800-897-5465. For more information regarding safety and to view available training resources, including helpful videos, visit safeu.utah.edu.

The Americans with Disabilities Act. The University of Utah seeks to provide equal access to its programs, services, and activities for people with disabilities. If you will need accommodations in this class, reasonable prior notice needs to be given to the Center for Disability Services, 162 Olpin Union Building, (801) 581-5020. CDS will work with you and the instructor to make arrangements for accommodations. All written information in this course can be made available in an alternative format with prior notification to the Center for Disability Services.

Addressing Sexual Misconduct: Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a Civil Rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information. If you or someone you know has been harassed or assaulted on the basis of your sex, including sexual orientation or gender identity/expression, you are encouraged to report it to the University's Title IX Coordinator; Director, Office of Equal Opportunity and Affirmative Action, 135 Park Building, 801-581-8365, or to the Office of the Dean of Students, 270 Union Building, 801-581-7066. For support and confidential consultation, contact the Center for Student Wellness, 426 SSB, 801-581-7776. To report to police, contact the Department of Public Safety, 801-585-2677(COPS).

Names/Pronouns. Class rosters are provided to the instructor with the student's legal name as well as "Preferred first name" (if previously entered by you in the Student Profile section of your CIS account, which managed can be managed at any time). While CIS refers to this as merely a preference, I will honor you by referring to you with the name and pronoun that feels best for you in class or on assignments. Please advise me of any name or pronoun changes so I can help create a learning environment in which you, your name, and your pronoun are respected. If you need any assistance or support, please reach out to the LGBT Resource Center https://lgbt.utah.edu/campus/faculty_resources.php

Lecture Timetable:

Note: This is a tentative outline and is subject to change

	Date	Lecture (Chapters)	Discussion worksheet due 10:30	Quiz due 12pm
	1/6/25	Welcome! 1.1–1.4; 2.1–2.3 Covalent/ ionic bonds; Formal charge; Lewis structures; Line-bond structures; Functional groups		
1	1/8/25	1.5–1.10 Hybridization; MO theory; VSEPR theory		
	1/10/25	1.10–1.13 VSEPR theory; Dipoles; Polarity; physical props; solubility		
	1/13/25	2.4–2.7 Formal charges on C; dash-wedge notation; resonance; shorthand		
2	1/15/25	2.8–2.12; Curved arrows; resonance	✓	
	1/17/25	2.13; 3.1–3.4 Lone pairs; Brønsted-Lowry Acids and Bases; *pKa*		1
	1/20/25	**MLK Jr. Day** (no class)		
3	1/22/25	3.5–3.8 Proton transfer, curved arrows; K_a; functional groups/ acidity	✓	
	1/24/25	3.9; 6.7–6.10 Lewis Acids and Bases; nucleophiles and electrophiles		2
	1/27/25	14.1–14.7 IR Spectroscopy; IR absorption bands; structure elucidation		
4	1/29/25	14.8–14.13; 14.16 Mass Spectrometry; fragmentation; structure/formula elucidation; Degrees of Unsaturation	✓	
	1/31/25	4.1–4.8 Alkanes: constitutional vs conformational isomers		3
	2/3/25	Review; 15.1–15.3 Intro to NMR		
5	2/5/25	15.4–15.5; 15.11–15.12 Proton and carbon NMR signals; chemical shift	✓	
	2/7/25	**Exam I**: Chp. 1.1–1.13; 2.1–2.13; 3.1–3.9; 4.1–4.8; 14.1–14.13; 14.16		
	2/10/25	15.6–15.10 NMR Splitting; coupling constants; complex splitting; integration		
6	2/12/25	4.9–4.15 Cycloalkanes; constitutional/ cis-trans stereoisomers; polycyclics	✓	
	2/14/25	5.1–5.6 Stereochemistry in nomenclature; optical activity; enantiomers		4
	2/17/25	**Presidents Day** (no class)		
7	2/19/25	5.8–5.9; 5.11 Chirality; Diastereomers E and Z; Configurational isomers	✓	
	2/21/25	6.1–6.6; 7.1 Gibbs Free energy; K_{eq}; Mechanisms: substitution vs elimination		5
	2/24/25	7.2–7.5 Mechanism: SN2; coordination/ heterolysis		
8	2/26/25	7.6–7.8 Mechanism: E2; Elec add/ elim; Shifts; driving force;	✓	
	2/28/25	7.9; 6.11 Mechanism SN1 & E1; carbocation rearrangement; type of carbon		6
	3/3/25	7.11 Competition; RDS; nucleophile/ base strength; leaving group		
9	3/5/25	**Exam II**: Cumulative, but emphasis on Chapters 15.1–15.12; 4.9–4.15; 5.1–5.9; 5.11; 6.1–6.11; 7.1–7.9	✓	
	3/7/25	7.13; 12.1–12.3 Alkyl halides and tosylates in synthesis		
	3/10/25			
	3/12/25	Spring Break: No Lectures or Discussions		
	3/14/25			
	3/17/25	12.9–12.10 Elimination and oxidation reactions of alcohols		
10	3/19/25	13.5–13.6; 13.8–13.10 Formation of ethers and epoxides; enantioselective epoxide ring-opening;		
	3/21/25	12.4; 12.6; LAH; Organometallic reagents; limitations of R$\ominus$	✓	7
	3/24/25	12.2; 12.7; Protection of alcohols; Formation of alcohols		
11	3/26/25	8.1–8.5 Electrophilic addition; hydrohalogenation; acid catalyzed hydration	✓	
	3/28/25	8.6–8.8 Oxymercuration-reduction; Hydroboration-oxidation; hydrogenation		8
	3/31/25	8.9–8.11 1,2-dihalides & halohydrins; X_2; dihydroxylation		
12	4/2/25	**Exam III**: Cumulative, but emphasis on Chapters 7.11; 7.13; 8.1–8.12; 12.1–12.7; 12.9–12.10; 13.5–13.6; 13.8–13.10	✓	
	4/4/25	8.12 and Review Oxidative cleavage		
	4/7/25	8.13–8.14 Predicting products and synthesis strategies		
13	4/9/25	9.1–9.4 Intro to alkynes; review acidity. mechanisms and preparation	✓	
	4/11/25	9.5–9.7 Alkyne reduction/ hydration/ hydrohalogenation		9

	4/14/25	9.8–9.10 Alkyne halogenation/ ozonolysis/ alkylation		
14	4/16/25	9.11: Synthesis Strategies	✓	
	4/18/25	11.1–11.3 Functional group transformation		10
	4/21/25	11.4–11.5 Retrosynthetic analysis: solvent/ stereochemistry/ percent yield		
	4/22/25	**Last Discussion**		
15	4/23/25	Last Discussion worksheet due at 10:30 on Gradescope	✓	
	4/25/24	**Final Exam** 8:00–10:00: Cumulative, Chapters 1–9; 11–15		

COVID-19 Considerations:

- Please do not come to class if you are experiencing Covid-19 or flu symptoms (fever, chills, new or worsening chronic cough, runny nose, sore throat, new loss of taste or smell, nausea, vomiting, diarrhea, fatigue, headache, muscle/body aches, or difficulty breathing/shortness of breath).
- Join the lecture via Zoom and participate via Poll Everywhere. The Zoom link is on Canvas. There is one Poll Everywhere link for this class associated with Prof. Tang.

Mental Health and Well-being:
Student Wellness: Personal concerns such as stress, anxiety, relationship difficulties, depression, cross-cultural differences, etc., can interfere with a student's ability to succeed and thrive. For helpful resources contact the Center for Student Wellness at www.wellness.utah.edu or 801-581-7776 or the Counseling Center https://counselingcenter.utah.edu or 801-581-6826.

Student Wellness Centers:

- (Main Campus Location) wellness@sa.utah.edu ECCLES STUDENT LIFE CENTER 1836 STUDENT LIFE WAY SUITE 2100 SLC, UT 84112 801-581-7776

Health education information, presentations, prevention training on various topics. Direct services are also provided such as HIV/STD testing and well coaching. https://wellness.utah.edu

- Victim Advocacy Location advocate@sa.utah.edu 201 SOUTH 1460 EAST STUDENT SERVICES BUILDING ROOM 328 SLC, UT 84112 801-581-7779

Support for survivors of sexual and relationship violence.

Counseling Center: Meets with students on a variety of personal and academic issues.
201 S 1460 E, Rm 426 Student Services Building 801-581-6826
Hours: Monday-Friday 8 a.m. - 5 p.m. Emergencies: For after-hours emergencies, contact the 24/7 Crisis Line 801-587-3000 https://counselingcenter.utah.edu

Women's Resource Center: Individual, Group, and Couples Counseling using a feminist multicultural framework. Services offered on a sliding scale fee.
Hours: Monday-Friday 8 a.m. - 5 p.m. University Union. Room 411. 801-581-6402
https://womenscenter.utah.edu

The Office of the Dean of Students: Excellent resource for students that are experiencing significant life challenges that are creating barriers to success. Please contact the office if you are experiencing significant hardship and would like to know your options. https://registrar.utah.edu/handbook/college-dean-contact.php Also, Behavioral Misconduct and Intervention. If you feel unsafe on campus or threatened report incidents through using https://deanofstudents.utah.edu/. University Union | Room 270 For emergencies, call Campus Police 801-585-2677 or 911.

Undocumented Student Support. Immigration is a complex phenomenon with broad impact—those who are directly affected by it, as well as those who are indirectly affected by their relationships with family members, friends, and loved ones. If your immigration status presents obstacles to engaging in specific activities or fulfilling specific course criteria, confidential arrangements may be requested from the Dream Center. **Arrangements with the Dream Center will not jeopardize your student status, your financial aid, or any other part of your residence.** The Dream Center offers a wide range of resources to support undocumented students (with and without DACA) as well as students from mixed-status families. **To learn more, please contact the Dream Center at 801.213.3697 or visit dream.utah.edu.**

Basic Needs, Food & Housing Security: Any student who faces challenges securing food or housing and believes this may affect course performance is urged to contact a Student Success Advocate for support at studentsuccess@utah.edu and see University resources at https://asuu.utah.edu/displaced-students or https://feedu.utah.edu/

Tutoring: Tutoring is available through the University of Utah Tutoring Center in the Student Services Building, Room 330. Students are given a list of tutors to contact and schedule for day, evening, or weekend appointments. Low-income students may qualify for free tutoring. For more information call 581-5153 or visit www.sa.utah.edu/Tutoring/. The Chemistry Department also has a help center for organic chemistry classes, located in room 2619 TBBC. The help center is usually staffed throughout the day, with specific hours posted on the door.

Veterans Center. If you are a student veteran, the U of Utah has a Veterans Support Center located in Room 161 in the Olpin Union Building. Hours: M-F 8-5pm. Please visit their website for more information about what support they offer, a list of ongoing events and links to outside resources: http://veteranscenter.utah.edu/. Please also let me know if you need any additional support in this class for any reason.

English Language Learners. If you are an English language learner, please be aware of several resources on campus that will support you with your language and writing development. These resources include: the Writing Center (http://writingcenter.utah.edu/); the Writing Program (http://writingprogram.utah.edu/); the English Language Institute (http://continue.utah.edu/eli/).